

2.4 GHz 5th-Order Chebyshev Stepped-Impedance LPF

Keysight ADS · Keysight LineCalc · CST 3D EM · FR4 Microstrip

Synthesized a distributed microstrip low-pass filter for ISM band avionics interference rejection. Conducted Monte Carlo yield analysis for fabrication tolerances and validated insertion/return loss in CST full-wave 3D simulation.

Two-Wheeled Inverted Pendulum (TWIP) Robot: LQR vs. Cascaded PID

Control Theory · MPU6050 · C++ Firmware · Complementary Filter

Formulated complete non-linear equations of motion and state-space matrices for a TWIP robot. Built physical prototype with real-time sensor fusion and cascaded PID stabilization, evaluating practical limitations against theoretical LQR.

CNC 2D Handwriting Machine & Vector Plotter

Arduino Uno · GRBL v1.1 · A4988 · NEMA 14 · G-Code Execution

Engineered a dual-axis electro-mechanical drafting plotter with custom-modeled CAD gantry and sub-millimeter precision. Configured GRBL firmware for high-speed synchronized stepper pulse coordination.

Autonomous Heavy Competition Rover

ESP32 Dual-Core · BTS7960 43A · 4S2P Li-Ion Pack · Telemetry

Designed high-torque rover featuring dual BTS7960 motor driver stages, JGA-25-370 metal geared motors, custom 16.8V Li-Ion battery pack, and real-time WiFi/BLE telemetry diagnostics.

 EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant — Embedded Systems & Controls

Jul 2025 – Present

Self-Employed · Remote

- Consulted on microcontroller firmware, sensor interfacing (I2C/SPI), and closed-loop algorithm synthesis.
- Conducted hardware debugging, circuit optimization, and component selection for international clients.

Head of Avionics Club — National University of Sciences & Technology (NUST)

2024 – Present

CAE, Risalpur, Pakistan

- Leading technical student initiatives, autonomous robotics workshops, and aerial telemetry testbeds.
- Organizing hands-on engineering hackathons bridging academic classroom theory with lab implementation.

A-Level STEM Tutor (Mathematics & Physics) — Private Mentorship

Nov 2022 – Present

Remote & In-Person

- Mentored students in advanced differential calculus, kinematics, wave mechanics, and electromagnetism.

 PROFESSIONAL CERTIFICATIONS

IBM Professional Certification: Data Analysis with Python — NumPy, Pandas, Scikit-Learn

IBM Verified Credential